

B.TECH FIRST YEAR PROJECT

School Administration Management System

Using Core Java & Object-Oriented Programming

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Session

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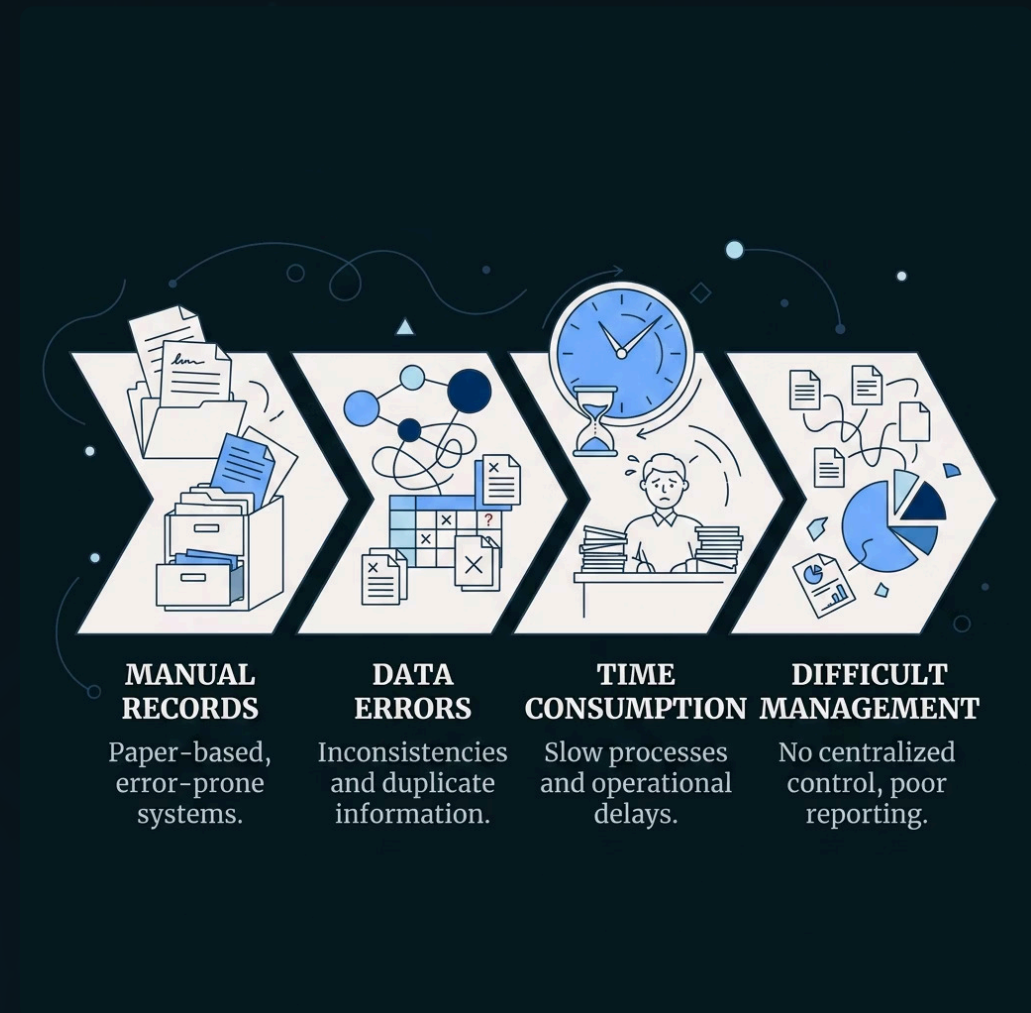
Introduction & Problem Statement

About the Project

A console-based **School Administration Management System** built entirely in **Core Java**, applying OOP principles to automate and streamline daily school operations — from student records to fee collection.

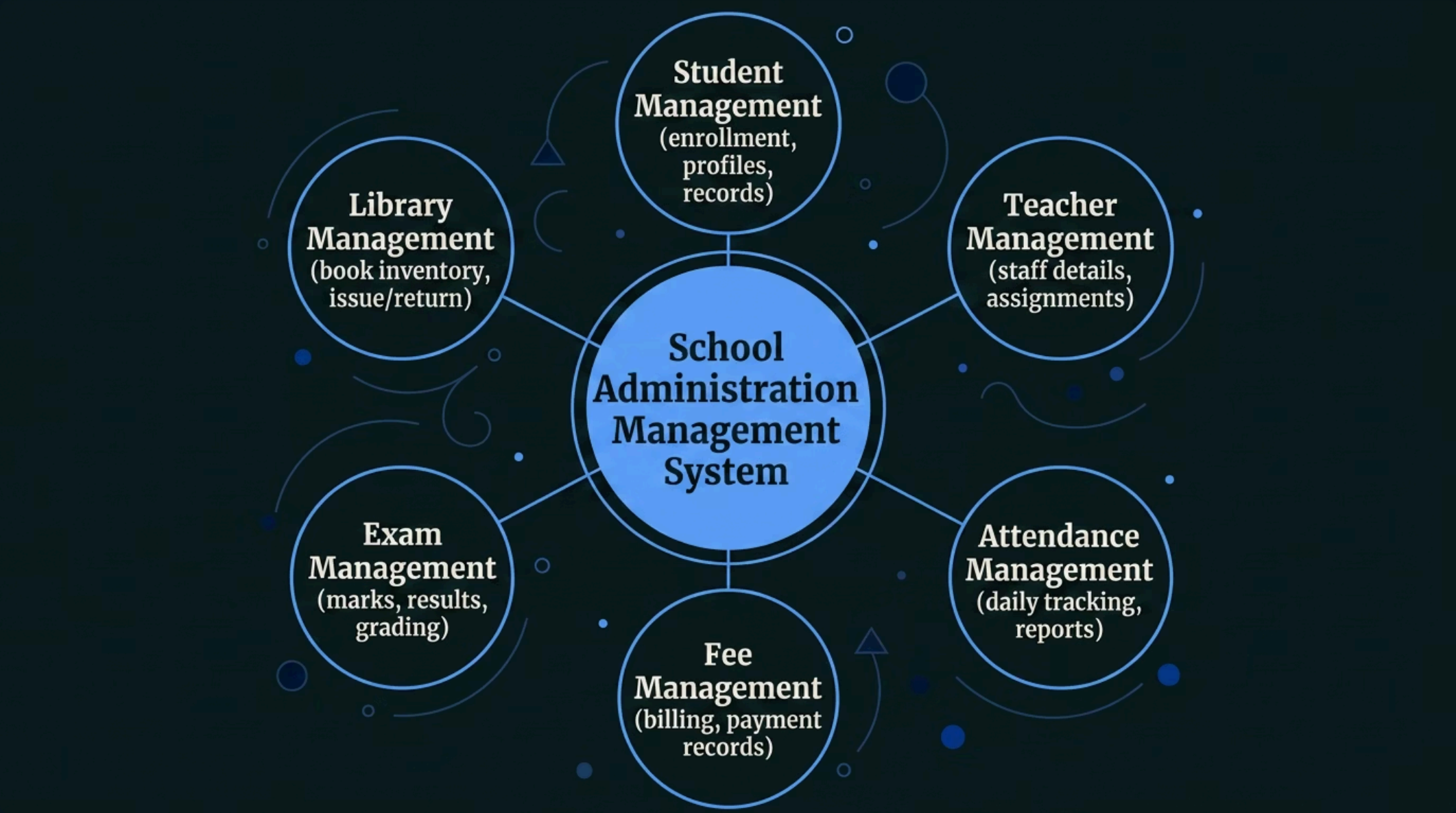
- Eliminates manual, paper-based record keeping
- Reduces human error in data management
- Provides a structured, menu-driven interface
- Enables fast retrieval and report generation

The Problem We Solve



Project Objectives

Six core modules work together to deliver a complete school administration solution.



Student Management

Enrollment & profiles



Teacher Management

Staff & assignments



Attendance

Daily tracking



Fee Management

Billing & payments



Exam Management

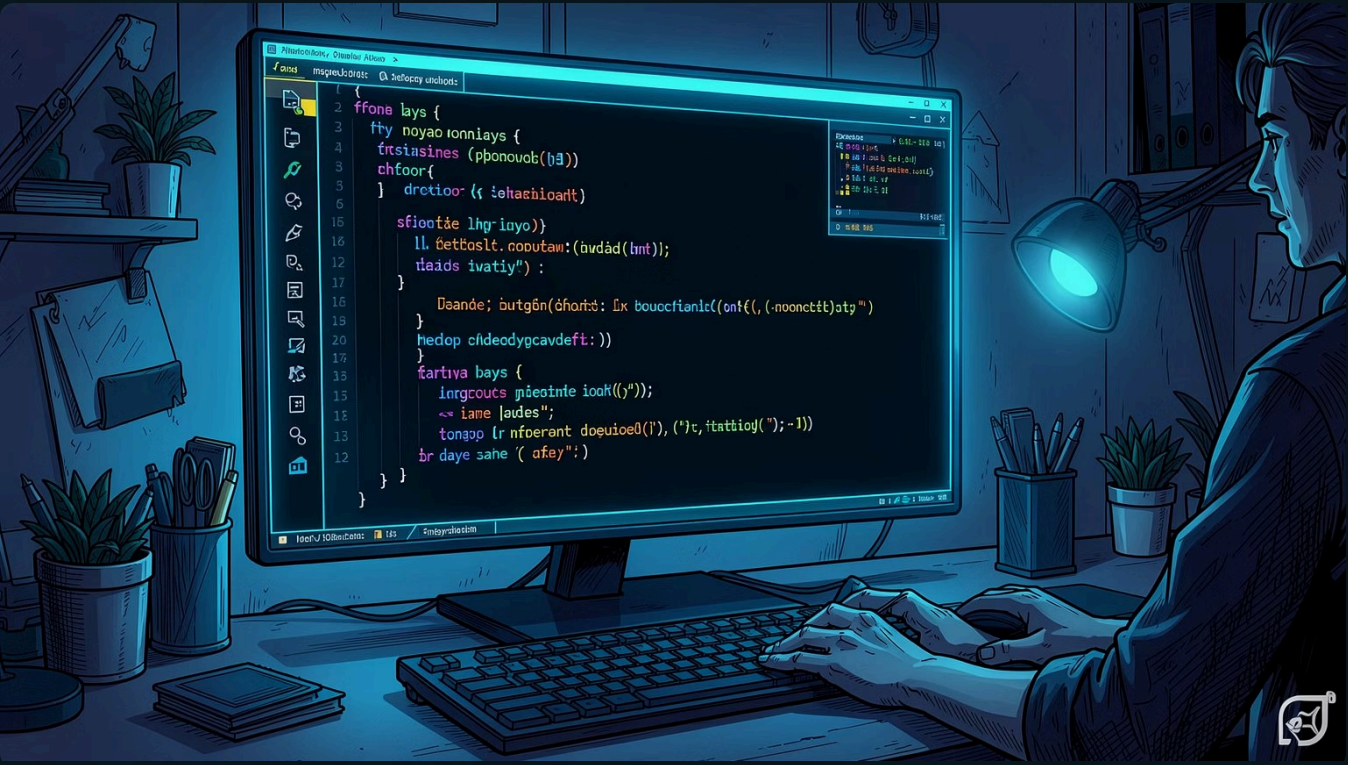
Marks & results



Library

Book issue/return

Technology Stack & OOP Concepts



Core Java Technologies

Core Java (JDK 8+)

Foundation language for all modules

Collections Framework

ArrayList, HashMap for dynamic data storage

File Handling

Reading/writing .txt and .dat files for persistence

Serialization

Saving and loading object states to disk

OOP Principles Applied



Encapsulation



Inheritance



Polymorphism



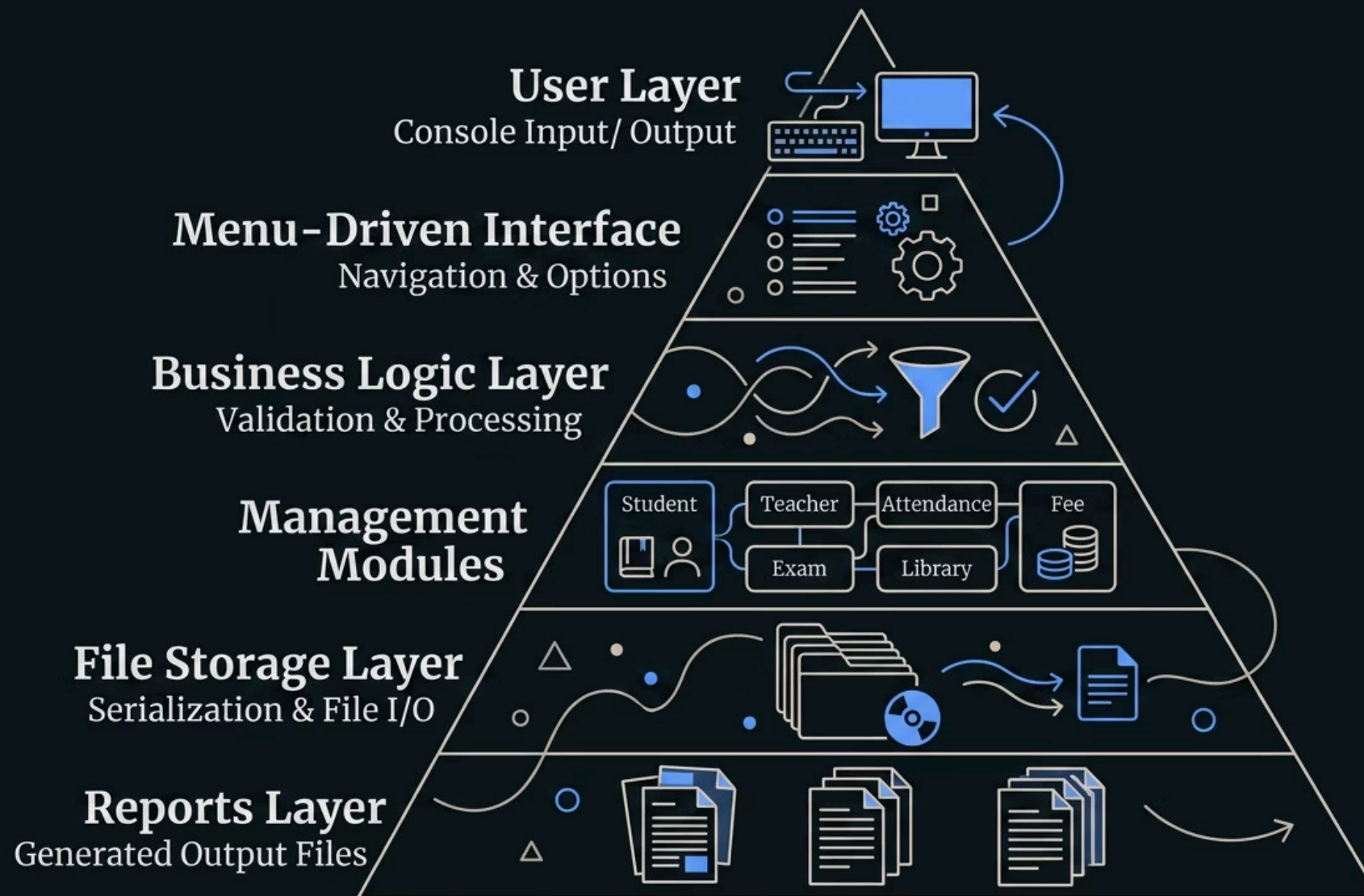
Abstraction



Exception Handling

System Architecture

A clean, layered architecture ensures separation of concerns — each layer has a single responsibility, making the system modular and maintainable.



User Layer

Console-based input and output interaction

Interface Layer

Menu-driven navigation and option selection

Business Logic

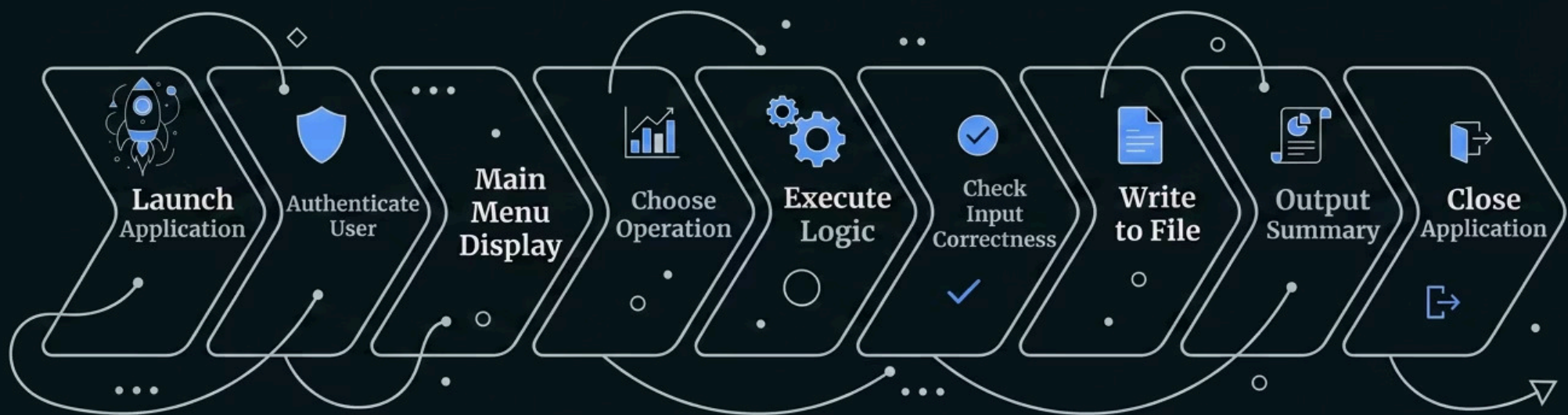
Validation, computation, and data processing

Storage Layer

File-based persistence using serialization

Working Flow

The system follows a structured, sequential workflow ensuring data integrity at every step of user interaction.



1

Start & Login

Authenticate user credentials

2

Dashboard

Display main menu options

3

Select Module

Choose operation type

4

Process & Validate

Execute logic, verify input

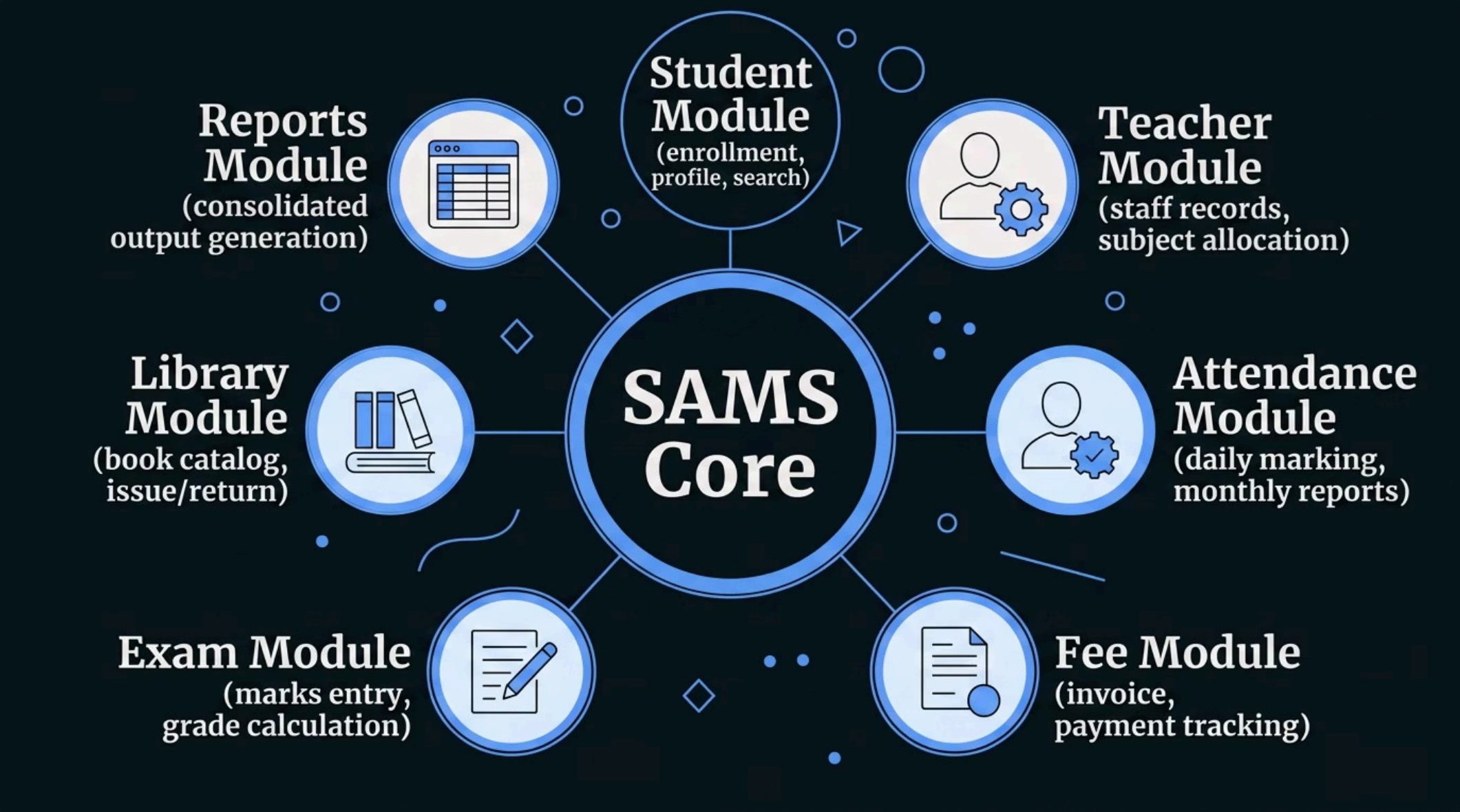
5

Save & Report

Persist data, generate output

Functional Modules

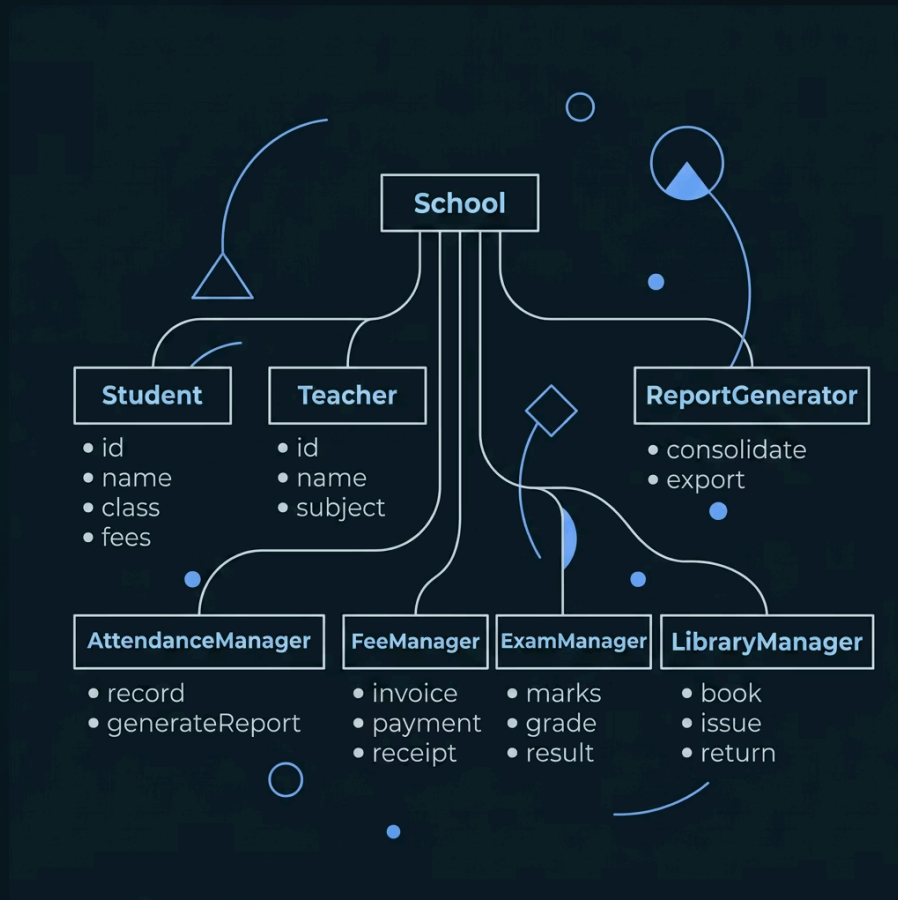
Seven interconnected modules form the backbone of the system, each encapsulated as an independent Java class with dedicated responsibilities.



		
Student Module	Teacher Module	Attendance Module
Enrollment, profiles, and search	Staff records and allocations	Daily marking and reports
		
Fee Module	Exam Module	Library Module
Invoicing and payment tracking	Marks entry and grading	Catalog and issue/return
		
Reports Module		
Consolidated output generation		

Class Diagram & Project Snapshots

UML Class Structure



Output Screens

```
Run: SchoolManagementSystem
adityakumar@AdityaKumars-MacBook-Air SchoolManagementSystem_01 % /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/adityakumar/Library/Application\ Support/Code/User/workspaceStorage/e55f42ddf29de41f160d1fa0a299beab/redhat.java/jdt_ws/SchoolManagementSystem_01\ _cbc5e2a4/bin SchoolManagementSystem

=====
SCHOOL ADMINISTRATION SYSTEM
=====
Username : Pankaj
Password : Pankaj123

===== MENU =====
1. Student Management
2. Teacher Management
3. Subject Management
4. Library Management
5. Attendance Management
6. Examination & Results
7. Fee Management
8. Notice Board
9. Reports
0. Exit
Enter Choice :
[Auto Save Completed]

[Auto Save Completed]
```



Main Menu

[Screenshot: Menu-driven navigation with module options]



Student Management

[Screenshot: Add/Search/Display student records]



Attendance Screen

[Screenshot: Daily attendance input and summary]

Additional Snapshots & Future Scope

More Project Outputs



Fee Management

[Screenshot: Fee invoice generation and payment status]



Library Management

[Screenshot: Book issue, return, and availability]



Result Generation

[Screenshot: Mark sheet and grade card output]

Roadmap Ahead

Current Version

1

Console-based Core Java

2

GUI Version

MySQL Integration

3

Java Swing / JavaFX

JDBC for database layer

4

Parent Portal

Cloud & Mobile

5

Web-based access for guardians

Cloud storage + Android app

Conclusion & Thank You



Process Automation

Automated core school operations, eliminating manual paperwork



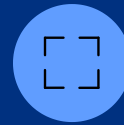
OOP Implementation

Demonstrated all four pillars of OOP in a real-world system



Efficient Records

Fast, accurate data management with file-based persistence



Scalable Design

Modular architecture ready for GUI, database, and cloud upgrades

- ✓ The **School Administration Management System** successfully demonstrates how Core Java and OOP principles can solve real-world administrative challenges — delivering a robust, maintainable, and extensible solution.

Thank You

Open for Questions & Answers